

SCHEDULE-ALIGNED IDENTITY-LOCK PERIODICITY UNDER AN ACTIVE-BLOCK RECORDER

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ABSTRACT

Position-decomposed diagnostics of diffusion large language models can confuse sampler geometry with model behavior because an active-block recorder exposes positions only through the current sampling window. We hold the model, questions, canvas, decoding rule, and total function evaluations fixed while applying a composite intervention: block partition changes together with the required per-block step allocation. The source manuscript reports that a registered candidate selector returns the configured length for one reused L32 development control, two main arms (L16/L64), and two sequential canvas-96 probes (L24/L48). An alternative rank-sum ordering computed from the same profiles gives the same reported choice; it is neither independent evidence nor calibrated inference. The L24 conditional margin is marginal ($p_{\text{margin}} = 7.1 \times 10^{-2}$), the L48 separation rests on one boundary locus, and the probes' complete candidate vectors are absent from the accessible package. Thus the probe argmaxes cannot be independently audited here. We recommend reporting separately what boundary-set composition makes reachable before data and what measured magnitudes support. The composite intervention, development reuse, sequential probes, post-treatment scope amendment, and untested cross-model, cross-schedule, and cross-recorder scope preclude a model-mechanism claim.

1 INTRODUCTION

Diffusion language models generate discrete sequences by iterative denoising (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024; Nie et al., 2025). Block diffusion provides the closest verified architectural context: it is autoregressive across blocks and applies discrete denoising within each block (Arriola et al., 2025). As decoding schedules become variable or structured (Li et al., 2025; Sun et al., 2026; Wang et al., 2026), a position-resolved diagnostic can inherit periodicity from the schedule that exposes its positions. Without an intervention, such a readout cannot distinguish model-intrinsic state, sampler state, and what the recorder makes observable.

We examine identity-lock time on a blockwise-decoded LLaDA-style model (Nie et al., 2025; 2026). An apparent period of 32 motivates a matched composite intervention over the block partition and its required per-block step allocation: the model, 300 questions, canvas, total NFE, decoding rule, and temperature remain fixed. The active-block recorder, however, logs only masked positions in the currently active block. Its lock-time profile is therefore defined relative to the sampling window (Figure 1).

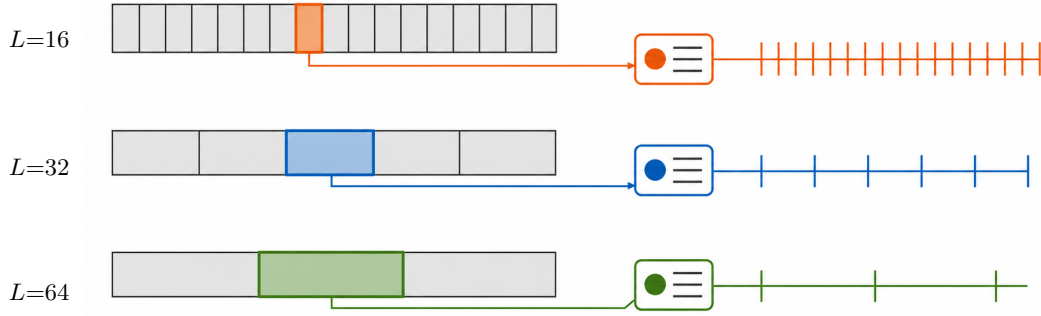


Figure 1: Qualitative measurement pipeline. From top to bottom, block length L grows while canvas extent stays fixed; color marks the one active block exposed to the recorder, and the right-hand ticks depict the boundary spacing visible through that window. Pulse and group counts are schematic, not quantitative evidence. Measured profiles and separations are in Figure 2 and Table 3.

Our evidence makes three scoped contributions. First, the source-reported point selector returns $P^* = L$ for two main arms, one reused development control, and two sequential probes. A rank-sum ordering of the same profiles gives the same reported choices, but is not an independent or inferential confirmation. This is descriptive selection-and-stability evidence rather than a null-calibrated period test. Second, the measured candidate-32 signs and separately registered non-power-of-two probes do not reveal a fixed-32 component at the recorder-observable layer; the missing complete probe vectors prevent an independent argmax audit, and the result does not rule out an unobserved model or sampler component. Third, this case study motivates a reporting recommendation: specify before data which outcomes boundary-set composition makes reachable, then report separately what the measured magnitudes show.

The evidential limitations are part of those contributions. The L24 corrected margin is marginal at $p_{\text{margin}} = 7.1 \times 10^{-2}$, the L48 estimate rests on one boundary locus, and the reused L32/NFE48 arm is a development control. Moreover, the originally registered model-intrinsic branch was withdrawn after treatment (amendment v3; Appendix A). The paper therefore audits a recorder-relative measurement on the tested beds; it does not establish a causal subtype or a mechanism across models, schedules, or recorders.

2 MEASUREMENT AND DESIGN

Core notation is defined here before it is used; Appendix C retains the longer terminology crosswalk.

Notation. For decode i and relative canvas position r , $\ell_{i,r}$ is the recorded identity-lock time. Its positionwise median is $m(r)$ and $d(r) = m(r+1) - m(r)$. For candidate period P , $\mathcal{B}_P$ contains candidate boundary indices and $\mathcal{I}_P$ their interior complement; $\text{sep}(P)$ is the boundary-minimum minus the interior-maximum of $d(r)$, and P^* maximizes it. “Canvas 96” denotes the separate 96-position probe bed. BINK denotes a restricted content null with k equally populated offset-rank strata.

Numerical provenance. Unless a quantity is explicitly introduced as a mathematical definition or an arithmetic identity, every empirical number, count, schedule, estimate, band, resampling summary, duration, and software version in this paper has status SR/CV: *source-reported and cannot be verified locally*. The accessible package contains the manuscript and two raster figures but not the event streams, analysis programs, complete registration records, or machine-readable results. No analysis was rerun for this revision. In particular, complete L24/L48 candidate-separation and rank-sum vectors are absent and are not reconstructed.

Model bed. We use an LLaDA-style masked dLLM at the LLaDA-8B-Instruct scale with a locally registered checkpoint on GSM8K (Cobbe et al., 2021). Decoding uses temperature 0 and

grid, define $\mathcal{B}_P = \{r \in \mathcal{D}_C : r \bmod P = P - 1\}$ and $\mathcal{I}_P = \mathcal{D}_C \setminus \mathcal{B}_P$. A candidate is admissible only when it belongs to that grid and both sets are nonempty. The registered estimator is

$$\text{sep}(P) = \min_{r \in \mathcal{B}_P} d(r) - \max_{r \in \mathcal{I}_P} d(r), \quad P^* = \arg \max_P \text{sep}(P). \quad (2)$$

Exact ties select the first (smallest) period in the ordered registered grid. Positive separation requires every candidate boundary jump to exceed every interior jump. On a strictly L -supported profile it penalizes divisors with false boundaries and multiples that leave true boundaries inside; without that condition it is not a general divisor/multiple guarantee. The registered selection-and-stability criterion fires only under the conjunction $\text{sep}(P^*) > 0$ and a cluster-bootstrap modal period equal to P^* with mode fraction at least 0.95. This operational criterion is not a null-calibrated hypothesis test: its bootstrap mode fraction measures stability under decode resampling, not a false-positive rate. The source protocol reports resampling the 300 decodes with replacement, preserving multiplicity, for $B=2,000$ replicates. Replicate-level outputs needed to reconstruct the criterion’s nesting are not in the accessible package.

Selection stability and its honest reading of “no misses.” Two resampling readouts are source-reported. First, a *joint paired cluster bootstrap* resamples a shared index set of the 300 decodes once and applies the same indices to the two main arms, the reused development control, and the two sequential canvas-96 probes. This is a paired summary of materially different evidence tiers, not five independent confirmations. Across $B=2,000$ replicates under a fixed opaque seed, $\Pr[P_{16}^*=16 \wedge P_{32}^*=32 \wedge P_{64}^*=64] = 1.0000$; the source also reports marginal $\Pr[P^*=L] = 1.0$ for each arm (Table 9). The exact item-pairing manifest and replicate indicators are absent. Second, Table 2 reports $\Pr[\text{sep}(L) > 0]$ rather than “0 misses.” The package does not resolve how this positivity indicator was nested with the separately computed modal-period fraction, so the table does not label it as a recomputed conjunction.

Integer lock times do *not* imply integer medians or separations. With an even sample such as $n = 300$, the conventional midpoint median can be half-integral; the reported L24/L48 values contain such half-steps. The exact median implementation is unavailable, so its convention cannot be audited. Under the conventional rule, $m(r)$, $d(r)$, and $\text{sep}(P)$ lie on a half-step lattice. This discreteness permits repeated values, but does not by itself force a bootstrap point mass. The collapsed L16 band is therefore an observed source-reported property of that bootstrap distribution, not a consequence of integer-valued medians. A 0/2,000 miss count is read only as sample-selection stability, never false-positive control.

Because sep is an extremum statistic, we also report a *non-extremal* alternative readout: a rank-sum comparison between candidate-boundary and candidate-interior values of the *same* $d(r)$ profiles. The source reports that its maximizing candidate is L for the two main arms, reused control, and two sequential probes. This is a same-profile sensitivity statistic, not independent evidence; its dependence, tie correction, normalization, and inferential calibration are unavailable. The values labeled “ z ” in the source are retained below only as uncalibrated standardized summaries. Complete candidate score vectors are absent, so their argmaxes cannot be audited from this package. A quantile-based separation survives only on L64 and L48 and selects multiples otherwise, which we report honestly as partial rather than as a full vote for the extremum statistic.

Replication accounting. Table 1 is the complete resampling inventory. Period selection and its interval share one draw, whereas the sign and stability checks use separate draws; Table 3 identifies the shared readouts. Uncounted values are observed-data point estimates; artifacts are in Table 9.

Matched intervention and scope. The two treatment arms and frozen development control are in Table 1. At fixed total NFE, changing the partition necessarily changes per-block step allocation, so the treatment is partition-plus-required-allocation. A source-reported within-arm rank transform preserves $P^*=L$ and each candidate-32 sign for the two main arms, reused control, and two sequential non-power-of-two probes $L=24$ and $L=48$ (Table 9); a uniform-allocation arm was not registered. The result therefore identifies a recorder-output association with the composite partition-plus-allocation schedule, not which component places or scales the boundary. A factorial allocation/partition intervention is future work.

Registration and two-layer reachability. The source manuscript states that the estimator and selection-and-stability criterion were fixed before treatment, but amendment v3 withdrew the model-intrinsic decision branch after treatment. The registration files and timestamps are absent, so this chronology cannot be independently audited; Appendix A, Table 8, retains only an opaque sequence. The canvas-96 probes were registered separately before their execution. We therefore report reachability at two layers. *A priori*, nesting such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ determines set composition but not the sign of $\text{sep}(32)$. *From events*, the observed magnitude ordering gives $\text{sep}(32) = -2$ on L16 and a negative value on L64. Thus a recorder-observable fixed-32 component is not selected under these active-block regimes; this does not identify model-intrinsic or sampler-state causes. The exact loci and synthetic condition are consolidated in Table 6; the legacy verdict labels remain only in Appendix C.

Separately conditioned content analysis. On the separate L32/NFE32 bed, BIN8 permutes confidence within eight equally-populated offset-rank bins after a fixed nuisance fit. The initial confidence conf0 is read at the first recorded step of a position’s block, following confidence-ordered parallel decoding (Ghazvininejad et al., 2019; Nie et al., 2025). This supporting diagnostic is not a cross-block-length validation.

3 RESULTS

3.1 THE REGISTERED SELECTOR FOLLOWS BLOCK LENGTH

The source reports that the reused development control recovers its known period and the two main arms move to their respective block lengths (Figure 2; Table 3). For these three canvas-256 arms, the printed complete four-candidate rows show that only the selected period has positive min–max separation. The two sequential probe argmaxes are also reported as their configured lengths, but the complete probe vectors are absent and cannot be independently checked. The paired main-arm/control relation is reported in all 2,000 shared-cluster resamples. A rank-sum ordering of the same profiles gives the same reported candidate in all evidence tiers; it is an alternative statistic, not independent calibrated inference.

The source-reported top-minus-second margins range from +2.0 to +8.0 d -units. Four unadjusted conditional-shuffle readouts have $p_{\text{margin}} \leq 2.1 \times 10^{-2}$, whereas L24 is marginal at $p_{\text{margin}} = 7.1 \times 10^{-2}$ (Table 5). These are descriptive diagnostics for a particular within-block shuffle construction, not significance or rejection claims, an arm-level false-positive-rate estimate, or family-wise calibration of the complete candidate-selection procedure.

Candidate 32 under the active-block recorder. Table 3 reports $\text{sep}(32) = -2$ on the measured $L = 16$ arm. The exact values in Table 6 show why: candidate-32 boundary values remain positive, but the measured odd-16 interior is larger. The joint sign $\text{sep}(16) > 0 \wedge \text{sep}(32) < 0$ holds in

Readout	L32		L64	L24	L48
	L16	control		probe	probe
$\Pr[\text{sep}(L) > 0]$, joint bootstrap	1.0	0.8805	0.9995	0.999	1.0
Reported rank-sum “ z ” at L (uncalibrated)	13.9	8.3	4.4	3.9	2.0
Wrong-candidate fire rate, original three-bed probe	0.0	0.0	0.0	—	—

Table 2: Source-reported selection stability and same-profile alternative readout (SR/CV; neither is a false-positive rate). The first row reports only the supplied positivity fraction; replicate-level data needed to reconstruct a nested two-part conjunction are unavailable. The L32 control’s 0.8805 is the weakest arm and appears alongside the source-reported modal-argmax relation. The second row retains the source’s standardized rank-sum label, but is not a calibrated z test and has no independent-evidence interpretation. The final row is a separate $B=1,000$ in-place wrong-candidate diagnostic for the original three beds: its zeros follow from negative point estimates and are not fresh-null false-positive rates. Within an arm, cells share draws. Availability is summarized in Table 9.

L (source)	$\text{sep}(P)$				P^* (mode frac.)	$\text{sep}(P^*)$ stability band [†]	$(\min_B, \max_I)$ @ P^*	$ B_{P^*} $	$ I_{P^*} $
	$P = 8$	$P = 16$	$P = 32$	$P = 64$					
16 (measured)	−1	+1	−2	−2	16 (1.0)	[+1, +1]	(+2.0, +1.0)	15	240
32 (control)	−1	−1	+1	−1	32 (1.0)	[0, +2]	(+2.0, +1.0)	7	248
64 (measured)	−1	−1	−1	+2	64 (1.0)	[+1, +3]	(+3.0, +1.0)	3	252
24 (probe)	—	—	—	—	24 (1.0)	[+1.5, +4.0]	(+5.0, +1.0)	3	92
48 (probe)	—	—	—	—	48 (1.0)	[+4.0, +6.5]	(+8.0, +1.5)	1	94

Table 3: Source-reported development-control, main-arm, and sequential-probe results (SR/CV). Each arm is reported with 2,000 cluster-bootstrap replicates; the mode fraction and percentile band are paired readouts of the same draw. The $(\min_B, \max_I)$ columns report the boundary minimum and interior maximum, followed by their observed set cardinalities. For the three canvas-256 rows, the four printed cells are the complete candidate grid. Dashes in the probe rows mean *not supplied*, not zero: their complete six-candidate vectors are absent, so the source-reported argmaxes cannot be independently audited. Half-integral values are compatible with an even-sample conventional median. The bands diagnose selection stability rather than coverage: L16 is reported as a point mass, while the L48 probe rests on one boundary location. Tables 1, 2, 5, 7, and 11 give the complete resampling inventory and separately budgeted checks; availability is in Table 9.

all 1,000 dedicated bootstrap replicates. Candidate 32 is therefore not selected at the recorder-observable layer on this bed. Nesting alone does not force that sign, and the result says nothing about an unobserved model-intrinsic or sampler-state component.

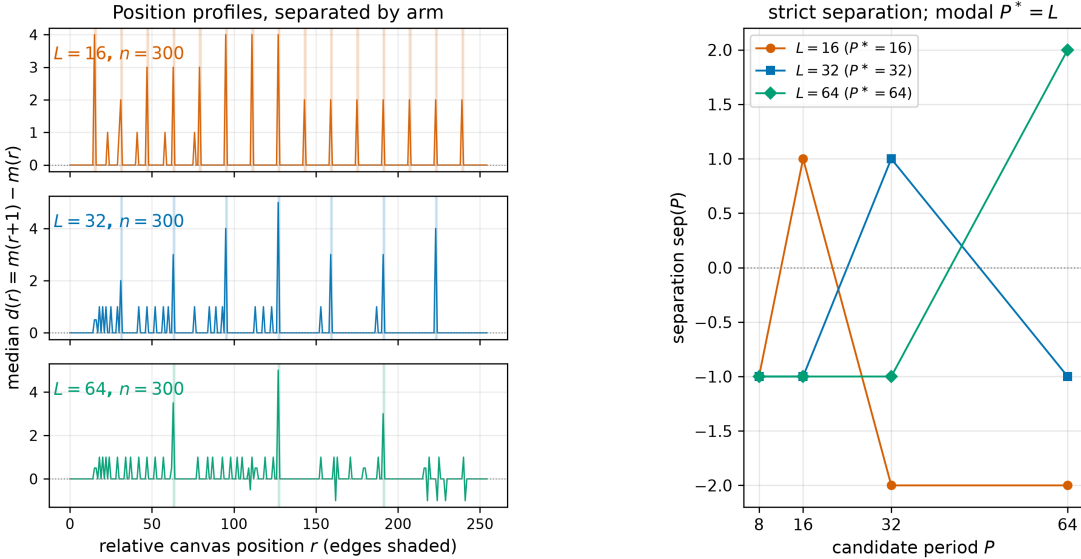


Figure 2: Pre-existing differential raster (all plotted measurements are SR/CV). Left: three stacked facets isolate the median adjacent-difference profile $d(r) = m(r+1) - m(r)$ for L16 (orange), the L32 frozen control (blue), and L64 (green), each from $n=300$ decodes; shaded ticks mark that arm’s block edges. The control is a reused contiguous development subset. Right: only the matching candidate has positive separation in each displayed arm. The source data and generating program are absent, so this figure was not regenerated for the revision and cannot be independently verified. The recorder-level comparison does not distinguish sampler from model mechanism.

Probe arm	Non-nested sep(32)	Other supplied cell
$L=24$ probe	-8.0	$P=48$: -1.0; band $[-3.0, 0.0]$
$L=48$ probe	-7.5	—

Table 4: Source-reported cells for the sequential canvas-96 probes (SR/CV). Candidate 32 is disjoint from the $L=24$ boundaries. The L24 marginal cell concerns the mismatched $P=48$ candidate rather than selected $P^*=24$; the displayed percentile band is a selection-stability diagnostic, not a coverage-valid interval. These are not complete candidate vectors: the omitted L24/L48 values and all-candidate rank-sum scores are absent, so neither probe argmax can be audited from the package.

Arm	P^*	sep(P^*)	m_{obs}	p_{sepL}	p_{margin}
L16 (canvas 256)	16	+1.0	+2.0	5.0×10^{-4}	1.5×10^{-3}
L64 (canvas 256)	64	+2.0	+3.0	5.0×10^{-4}	1.0×10^{-3}
L24 (canvas 96)	24	+4.0	+5.0	5.0×10^{-4}	7.1×10^{-2}
L48 (canvas 96)	48	+6.5	+8.0	2.3×10^{-2}	2.1×10^{-2}
L32 control (canvas 256, NFE 48)	32	+1.0	+2.0	5.0×10^{-4}	1.0×10^{-3}

Table 5: Source-reported conditional separation-margin diagnostic (SR/CV; not full-selector calibration). Each of $R=2,000$ replicates under an opaque fixed seed permutes $d(r)$ within each true L -block, including its edge index, then records the argmax candidate’s top-minus-second gap. Here $m_{\text{obs}} = \text{sep}(P^*) - \max_{c \neq P^*} \text{sep}(c)$ and the discrete exact $p = (1 + \#\{m_{\text{perm}} \geq m_{\text{obs}}\})/(R+1)$. The p_{sepL} column tests $\text{sep}(P^*)$ alone; p_{margin} is the selection-controlled readout. Provenance and the original-gate size audit are in Table 9 and Appendix A.5.

Measured class	Loci	$d(r)$ values	Readout
L32 true-32 boundaries	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 5, 3, 3, 4]	development control
L16 true-16 odd chain	[15, 47, 79, 111, 143, 175, 207, 239]	[4, 3, 3, 4, 2, 2, 2, 2]	candidate-32 interior max 4
L16 would-be $\mathcal{B}_{32}$	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 4, 2, 2, 2]	boundary min 2; sep(32) = -2
L64 mid-block 32	[31, 95, 159, 223]	L64: [0, 0, 0, 0]; L32: [2, 4, 3, 4]	recorder-observable collapse
L64 true-64	[63, 127, 191]	L64: [3.5, 5, 3]; L32: [3, 5, 3]	true boundary remains
Strict-support synthetic L			
	16	{8, 16, 32, 48}	16 for every source
	32	{8, 16, 32}	32 for every source
	64	{8, 16, 32}	64 for every source

Table 6: Consolidated negative evidence, reachability, and development check. A priori, nested boundary sets determine membership but do not force the sign. From events, L16 candidate 32 has boundary minimum 2 and interior maximum 4; the joint sign holds in 1,000/1,000 bootstrap replicates. On L64, would-be 32 boundaries are recorder-observable interiors. The lower panel is a development-only synthetic control: $P^* = L$ holds for the shown strictly L -supported profiles, not for arbitrary residuals. The upper rows are observed-data readouts from their arm samples, not additional replications. Provenance is in Table 9.

3.2 SELECTION STABILITY, NOT NULL CALIBRATION OR MECHANISM

The central estimand is discrete period identity. On each supplied profile, positive min-max separation is a point-estimate property; the source-reported bootstrap probes stability to resampling

decodes. One shared question-index draw is reported across the two main arms, reused development control, and two sequential probes (Section 2); this pairing does not make them equivalent confirmations. The reported positivity fractions range from 0.88 to 1.0, with the reused L32 development control weakest at $\Pr[\text{sep}(32) > 0] = 0.8805$. The alternative rank-sum ordering uses those same profiles and has no independent or calibrated-inference status.

The evidence tiers also differ in support: L24 is marginal under the unadjusted conditional margin diagnostic, and L48 has $|B_{48}|=1$, so its positive separation is a single-locus observation. The L32 control reuses a pre-existing contiguous pool and its percentile stability band $[0, +2]$ touches zero (it is not a coverage-valid CI). Complete probe candidate vectors are unavailable. These limitations preclude a uniform five-confirmation reading, an arbitrarily small effect claim, or a universal period. A fresh in-session L32 control and additional probe loci remain unexecuted future work.

Exploratory spectral residual check. Because sep cannot exclude every residual shape for a nested candidate, a post-data DFT check tests one sinusoidal residual-32 family on L16. Its injected-residual control fires and its no-residual control is silent; the real L16 profile does not cross the family-wise threshold at $P = 32$, and a direct boundary-subset test finds no amplitude split. This is a narrow negative for the tested family, not evidence that arbitrary residual shapes are absent. The estimator, complete candidate grid, power sweep, and thresholds are reported in Appendix A.1 and Table 7.

Canvas-96 non-nested probe (boundary disclosure). On the separate canvas-96 probe beds, candidate 32 is *non-nested* ($\mathcal{B}_{32} \cap \mathcal{B}_{24} = \emptyset$), and the same estimator does not select a recorder-observable fixed-32 component without relying on the canvas-256 nesting relation. Table 3 reports the two probe estimates, their selected block lengths, and the disclosed marginal mismatched cell (Table 4); their complete vectors are absent, and the source reports about 0.38 GPU-h. This is a scoped sequential extension (canvas 96 $\neq$ 256), not a replacement for the matched canvas-256 arms.

The L64 mid-block reversal and its L32 comparator are reported without a separate verdict vocabulary in the consolidated negative-evidence Table 6.

Supporting analysis outside the headline claim. A separately budgeted L32/NFE32 bed shows a moderate-granularity association between initial confidence and later lock ordering. Because it is not a cross-block-length validation and does not identify a schedule mechanism, its estimand, results, and null-granularity audit are reported in Appendix A.7.

4 LIMITATIONS

- **Observation is not mechanism.** The active-block recorder cannot separate model-intrinsic state, sampler state, and recorder-observable state. Candidate 32 is not selected only at the last layer under the tested regimes.
- **Registration history.** The estimator was selected with a failed development control before treatment, but the model-intrinsic decision branch was withdrawn after treatment. The non-nested probes have their own pre-run registration and a different 96-position canvas.
- **Bed and grid.** Evidence covers one model family, one task, and the candidate grids in Table 1; off-grid periods, other models, tasks, samplers, and untested block lengths are not covered.
- **Composite intervention.** Partition and required per-block step allocation vary together. Boundary location is measured; a distinct allocation effect on amplitude is not.
- **Boundary cardinality.** The min-max separation is an extremum statistic, so its support matters: on the probe arms the selected period rests on few measured boundary positions (the L48 probe has $|B_{48}|=1$), making its positive separation a single-locus readout rather than a stable margin. We report $|B_{P^*}|$ and $|I_{P^*}|$ per arm in Table 3 for this reason.
- **Probe auditability.** The current package omits the complete L24/L48 candidate-separation vectors and all-candidate rank-sum scores. The two probe argmaxes and claims about their unprinted candidates are therefore source-reported and cannot be independently audited.

- **Event semantics.** The event schema and recorder implementation are absent. The ordering domain $\mathcal{T}_{i,r}$ is finite, but whether its labels are global, block-local, or per-position and how missing, censored, or terminal cases enter $m(r)$ remain unresolved.
- **Recorder and control arms.** A dual-recorder comparison that logs the same trajectory through active-block and full-canvas/passive instruments is GPU-gated future work; no cross-recorder result is claimed. A fresh in-session L32/NFE48 run is also GPU-gated: the present control reuses a pre-existing contiguous pool, has the lowest positivity (0.8805), and its percentile stability band touches zero.
- **Content scope.** The content association comes from one L32/NFE32 bed and survives only through moderate null granularity (it is not established at the finest registered BIN32 granularity). Its behavior on the L16/L64 beds is not measured.

5 RELATED WORK

Foundational discrete-diffusion work supplies modeling background (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024). The nearest verified architectural source is Arriola et al. (2025), which defines block diffusion as autoregressive across blocks with discrete denoising inside each block. It establishes the blockwise decoding context, but does not study active-recorder identity-lock times or the finite-grid selector used here.

Two checked measurement sources motivate only a narrow confound comparison. Sahoo et al. (2026) shows that apparent linear-probe separation can be explained by task-format confounding after residualization; it does not analyze diffusion recorder windows. Feng et al. (2024) studies propositional probes for latent world states and motivates distinguishing an internal-state measurement from model outputs; it does not perturb an active-block observation window. Table 13 records these verified boundaries. Within this checked set, the present paper examines a different measurement object. This is not a priority or field-wide absence claim: the bibliography-wide metadata and sentence-level support audit remains incomplete. The post-treatment amendment, strictly- L -supported boundary, and tested-family-only DFT check remain explicit.

Reproducibility statement. Reproducing the reported analyses requires the listed event streams, analysis scripts, and private model checkpoint. The current submission source package does not include them, so no numeric rebuild was run from that package.

6 CONCLUSION

The source manuscript reports that the registered selector returns the configured block length for two main arms, one reused development control, and two sequential probes. This is not five equivalent confirmations: L24 is marginal under the conditional-shuffle diagnostic, L48 is a single-locus readout, and the probes’ complete candidate vectors are unavailable. Bootstrap summaries describe sample stability, while the rank-sum ordering is an alternative statistic on the same profiles rather than independent calibrated inference. The practical reporting recommendation is to state what set composition makes reachable before data and then separately state what measured magnitudes show. The composite partition-plus-allocation intervention supports no model, sampler, allocation, or recorder mechanism claim.

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A PROTOCOL CHRONOLOGY, MEASUREMENT CANON, AND SENSITIVITIES

A.1 EXPLORATORY SPECTRAL RESIDUAL CHECK

For a nested candidate such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$, a *negative* $\text{sep}(32)$ is not evidence of absence: the extremum comparison cannot recover every possible residual shape. An *exploratory* (post-data, not pre-registered) check therefore asks whether a *non-nesting* estimator recovers a residual source period that sep misses. We use a DFT periodogram: for each candidate P , $G(P) = \|\sum_r d(r)e^{-2\pi ir/P}\|/\|d\|$ is the normalized spectral magnitude of the adjacent-difference profile at frequency $1/P$, with a permutation null (shuffle $d(r)$ positions; 4,000 replicates) and family-wise q_{95} over all candidates. Harmonic testing follows the periodogram tradition (Fisher, 1929); our finite-grid threshold is a bespoke single-step maximum over candidates, using the standard permutation max-statistic construction (Nichols & Holmes, 2002). We also run a direct two-sample test of the two $L=16$ -boundary subsets in Table 6 (the odd-16 chain vs. the would-be- $\mathcal{B}_{32}$ positions), a reachable test of residual 32-amplitude modulation.

Case	$G(8)$	$G(16)$	$G(24)$	$G(32)$	$G(48)$	$G(64)$	$q_{95}@32$
Synthetic, residual-32 present	3.58	3.58	—	4.86	—	—	1.74
Synthetic, no residual	4.01	4.01	—	0.25	—	—	1.75
Real L16 measured	3.79	3.70	0.15	0.23	0.36	0.18	2.15 (FW)

Supporting check	Statistic	Threshold / reference	Reading
Injected residual	Residual-32 amplitude 0.5 over comb-16 amplitude 3.0; $G(32)=4.86$	$q_{95}=1.74$	fires
No-residual synthetic	$G(32)=0.25$	$q_{95}=1.75$	null
Real L16 remaining grid	$G(24)=0.15$, $G(48)=0.36$, $G(64)=0.18$	per-candidate $q_{95} \approx 1.71\text{--}1.78$	no residual candidate
Comb harmonic	$G(8)/G(16)=1.000$ (pure comb), 1.024 (real L16)	1.0 reference ratio	recorder-comb harmonic
Power sweep	Amplitude 0.15: no fire; 0.20: fire; grid step 0.05 (5.0%–6.7% of comb amplitude 3.0)	q_{95} above	transition bracketed
Odd-16 vs. would-be- $\mathcal{B}_{32}$	Mean $ \Delta d =0.036$; permutation $p=1.0$	$\alpha = 0.05$	no amplitude split

Table 7: Spectral (DFT) periodogram of the adjacent-difference profile $d(r)$ — an *exploratory*, non-nesting check (Section A.1). $G(P)$ is the normalized spectral magnitude at frequency $1/P$; “fires” means the 32-component exceeds its permutation q_{95} (4,000 replicates). The upper panel completes the synthetic diagnostic grid and reports every real candidate; the lower panel separates the supporting power, harmonic, and two-sample checks from their interpretation. The synthetic residual fires, whereas the comb-only synthetic and real L16 profile are null at the target period. On real L16, no candidate in the tested grid is significant; only the recorder-comb harmonics fire. The power sweep brackets sensitivity for the tested sinusoidal family only, and the direct boundary-subset comparison finds no split. Provenance is in Table 9.

Development control. On the recorder-consistent synthetic grid, with a clean sinusoidal residual-32 component overlaid on the $L=16$ comb, sep returns $P^*=16$ (blind) while the spectral 32-component fires; on the no-residual synthetic comb, both are silent (Table 7). This is a positive control for the tested sinusoidal family, not validation against arbitrary residual shapes.

Real L16 arm. The tested candidate grid does not exceed its family-wise threshold at $P = 32$, and the boundary-subset test finds no amplitude split. The registered power sweep brackets the tested-family transition and records its amplitude range in Table 7. Hence the measured arm does not trigger for this tested sinusoidal family and amplitude range; arbitrary residual shapes are not ruled out (Table 9).

A.2 REGISTRATION CHRONOLOGY

Opaque stage	Relation to data	Source-reported event
R0	Pre-main-data	Original registration preceded the main-arm execution.
R1	Pre-main-data	A development amendment replaced a failed signed mean-difference score with the min-max estimator and recorded the selection-and-stability criterion.
R2	Main-data	The two main arms were executed.
R3	Post-main-data	A scope amendment withdrew the model-intrinsic branch.
R4	Pre-probe-data	A separate non-nested canvas-96 protocol and its amendments preceded the two sequential probes.

Table 8: Opaque registration sequence (SR/CV). Raw timestamps and searchable internal identifiers are omitted from blind prose. The underlying registrations, trusted timestamps, and diffs are absent from the accessible package, so the sequence is source-reported and cannot be audited. It documents a post-main-data scope change, not a pre-main-data mechanism verdict.

A.3 OPAQUE EVIDENCE INDEX

Evidence ID	Role	Accessible-package status
A31-E1	Two main-arm and reused-control selector profiles	SR/CV; numeric targets and programs absent
A31-E2	Paired/marginal bootstrap and stability summaries	SR/CV; pairing manifest and replicate indicators absent
A31-E3	Sequential L24/L48 probes	SR/CV; complete candidate and rank-sum vectors absent
A31-E4	Conditional margin, rank-transform, and spectral checks	SR/CV; result arrays and programs absent
A31-E5	Separate content-association bed	SR/CV; events and analysis programs absent
A31-R1	Registration and amendment sequence	SR/CV; timestamped records and diffs absent
A31-V1	Figure 2	Raster present; source data and generator absent

Table 9: Blind-safe claim-to-evidence index. Opaque paper-local IDs replace raw workspace paths, timestamp strings, and digest or commit prefixes. The private crosswalk is unavailable in the current package; this table is an availability disclosure, not evidence that the source-reported numbers were reproduced.

A.4 MEASUREMENT CANON

For blockwise decoding observed through an active-block recorder, a position-resolved analysis should:

1. record block length, per-block steps, and generated-canvas coordinates;
2. calibrate within-block windows to the generating run rather than a fixed model constant;
3. report reachability twice: *a priori*, what estimator algebra and boundary-set composition imply; *from events*, what the measured magnitude ordering implies. In particular, $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ does not force $\text{sep}(32) \leq 0$;
4. perturb the schedule when attributing a position-resolved diagnostic;
5. test content claims against a structure-preserving null and report its granularity; and
6. mark cross-bed relationships unmeasured until the corresponding intervention is run.

This is a measurement canon, not a mechanism claim. The registered estimator returns $P^* = L$ on the displayed *strictly* L -supported profiles: non-degenerate jumps occur on all L boundaries and no L -multiple's exclusive-boundary residual outcores the L -boundary minimum. It is not a guarantee for arbitrary residual profiles, and identifying commit internals would require instrumentation of sampler-state transfer.

A.5 SELECTIVITY DISCLOSURE (PERMUTATION-GATE SIZE AUDIT)

Original-gate size. The source reports that the original within-block threshold was anti-conservative across the two main arms, reused development control, and two sequential probes: $10/25 = 40\%$ of wrong candidates (non-true- L grid points) were flagged by the original gate, so the nominal family-wise error of at most 5% was not realized. Table 10 gives the per-arm audit. The cause is that edge-including within-block permutation destroys the non-target global level/trend structure of $d(r)$, shifting the null downward and loosening the threshold.

Corrected margin and cross-condition red-team. Table 5 replaces that level gate with the top-minus-second separation margin and a discrete exact p . As a known-clean red-team, globally (cross-block) shuffling the same $d(r)$ constructs a structureless reference. The corresponding probabilities are consolidated in Table 10: four unadjusted cross-condition checks are below 5% and the sequential L24 probe is again marginal. This source-reported pattern is consistent with schedule-aligned structure rather than within-profile jitter, but it does not calibrate the full selector. A genuinely clean arm has no blocks to permute, however, so this red-team is a cross-condition comparison, not a per-arm data-shuffle false-positive rate.

Arm	Original-gate wrong candidates	Cross-condition p_{margin}
L32 (reused development)	3/5	1.5×10^{-3}
L16 (main)	4/5	5.0×10^{-4}
L64 (main)	3/5	2.0×10^{-3}
L24 (sequential probe)	0/5	5.5×10^{-2}
L48 (sequential probe)	0/5	1.8×10^{-2}
Aggregate	10/25 (40%)	4/5 below 5%

Table 10: Source-reported selectivity-gate audit (SR/CV), stratified as one reused development control, two main arms, and two sequential probes. The original threshold flags 10/25 wrong candidates. Four unadjusted cross-condition readouts are below 5%; L24 is marginal. These probabilities are not per-arm data-shuffle false-positive rates or full-selector calibration.

Reporting rule. Any threshold-type permutation gate used here for selectivity reports three quantities together: the wrong-candidate hit rate, a known-clean-arm or explicit cross-condition check, and the discrete exact p . The 10/25 audit and the margin correction therefore accompany every selectivity claim based on this within-block gate.

A.6 THREE-PART EVIDENCE DECOMPOSITION

For auditability, we separate the recorder-relative object $d(r)$, the pre-treatment min-max decision procedure, and the scoped empirical finding that P^* follows L on the tested recorder beds. The shared-cluster bootstrap probes sample stability, and the same-profile rank-sum alternative probes statistic sensitivity; the sequential non-nested canvas-96 probes bound the empirical finding but have incomplete candidate vectors. Table 9 maps each component to its availability status. This decomposition is a falsification aid, not three independent novelty claims or a mechanism attribution.

A.7 SUPPORTING CONTENT ASSOCIATION ON A SEPARATE BED

Estimand τ . On the separate L32/NFE32 bed, one unit is a decode-block with resolved positions $\mathcal{R}$; c_r is initial confidence and $o_r = r \bmod 32$ is block offset. The estimand is the median per-unit

rank association after residualizing confidence on offset:

$$r_u = \frac{\sum_{p < q \in \mathcal{R}: \ell_p \neq \ell_q} \text{sgn}(\ell_q - \ell_p) \text{sgn}(\tilde{c}_p - \tilde{c}_q)}{\#\{p < q \in \mathcal{R} : \ell_p \neq \ell_q\}}, \quad \tilde{c}_r = \text{rank}(\text{rank}(c_r) - \hat{\beta} \text{rank}(o_r)), \quad (3)$$

Here the registered statistic fits $\hat{\beta}$ once on the observed data and shuffles the resulting $\tilde{c}$ within offset-rank strata. A sensitivity check instead refits $\hat{\beta}$ inside every permutation replicate. Earlier lock with higher residualized confidence contributes +1. Inference is rooted at decode clusters: the 500-replicate interval resamples decodes and carries their block units with multiplicity. The position-fake keeps the observed amplitude but reassigns it to positions, destroying the observed content–position pairing. This differently budgeted bed supplies supporting, not cross-block-length, evidence.

Quantity	Estimate / band	Null q_{95}	Null readout
Content-layer τ_{res}	0.2722 [0.2649, 0.2812]	BIN16 0.20235	0/2000 exceed
Refit- $\hat{\beta}$ sensitivity	0.2722	BIN8 0.13545	0/2000 exceed
Same-amplitude position fake	max. margin +0.00979	observed margin +0.12837	below observed

Table 11: Content-versus-structure result on the separate $L = 32$, NFE-32 bed. The statistic τ is the median decode-block rank association between earlier lock and higher confidence after offset residualization. BIN16 is the load-bearing conservative, non-degenerate null; the per-replicate nuisance refit is a separate BIN8 sensitivity and replays bit-exactly. Table 12 reports the full granularity ladder and three-bucket one-flip classification. This supporting analysis is not a cross-block-length experiment.

The full-granularity ladder in Table 12 is essential. The observed association clears every non-degenerate restricted null through BIN16, but matches the zero-power point mass at BIN32. Thus content association is established at moderate granularity, never at the finest registered granularity; the construction-specific BIN32 line is not interpreted as a lower bound. Its behavior under $L \in \{16, 64\}$ is unmeasured, and the two analyses are not claimed to be mutually calibrated.

A.8 CONTENT ACCOUNTING AND NULL GRANULARITY

Accounting / sensitivity check	Recorded value	Interpretation
Content-resolved block-units	6,178 of 6,274 candidates	Terminal-resolved units excluded
Resolved comparable pairs	1,571,854	Both positions resolve
Within-unit support	Median 4 distinct locks; median resolved-pair fraction 0.575	Unit-level order is observed
Resampling	$n_{\text{boot}}=500$; $R_{\text{null}}=2000$	Decode-cluster interval and restricted null
K-ladder restricted-null q_{95}	$K=1$: 0.00486; BIN4 0.09276; BIN8 0.14384; BIN16 0.20235; BIN32 0.27221 (point mass)	With $\tau_{\text{res}}=0.2722$, margins at $K=1$, BIN4, BIN8, and BIN16 are 0.26735; 0.17945; 0.12837; 0.06986; BIN32 is zero-power by construction and carries no lower-bound reading
Nuisance- $\hat{\beta}$ sensitivity	Refit inside each BIN8 replicate: $q_{95}=0.13545$; fixed-fit BIN8 $q_{95}=0.14384$	0/2000 exceedances; bit-exact replay. BIN16 remains the load-bearing conservative non-degenerate arm
One-flip sensitivity ($1/n_{\text{selected}}$)	$n_{\text{selected}}=6178$; all 6178 residual-sign flips give $\Delta=0$	DEGENERATE, by construction; no pass claim. The null-specification ladder, not this identity arm, carries the sensitivity conclusion

Table 12: Unit accounting and restricted-null sensitivity for Table 11. A block-unit is rooted in its decode cluster; decodes are resampled with multiplicity. The registered line fits the nuisance slope once; the displayed sensitivity refits it inside each replicate. One-flip granularity is $1/n_{\text{selected}}$, not permutation spacing $1/R$. Under the three-bucket contract, maximum single-point effect $\phi = 0$ is DEGENERATE (by construction) and carries no pass/fail claim; $0 < \phi \leq \text{margin}$ would pass and $\phi > \text{margin}$ would fail. BIN32 is likewise zero-power by construction. The full sweep therefore limits the positive finding to moderate granularity. Provenance is in Table 9.

A.9 CLOSEST-WORK SETUP COMPARISON

Verified source	Supported local use	Boundary relative to this paper
Arriola et al. (2025)	Block diffusion is autoregressive across blocks and uses discrete denoising within blocks.	Architectural context; no active-recorder lock-time or period-selector study.
Sahoo et al. (2026)	Task format can confound a linear probe’s apparent reasoning-mode separation after residualization.	Instrumentation-confound caution; no diffusion recorder-window perturbation.
Feng et al. (2024)	Propositional probes monitor latent world states and distinguish internal measurements from outputs.	Measurement-layer context; no active-block observation intervention.

Table 13: Narrow closest-work matrix restricted to the three sources with primary-source support verified for these local sentences. The rows do not establish priority or a field-wide absence; the bibliography-wide audit remains incomplete.

B REPRODUCIBILITY AND ARTIFACT MAP

Execution cost, shard completion, and event-row validation are reported in Table 14, separately from provenance, so that spend is never conflated with a completeness check. Table 9 uses opaque paper-local evidence IDs. The current package contains the manuscript, bibliography, and two figures, but not the event, result, program, registration, or evidence bundle; the table is therefore an availability index rather than a self-contained artifact release.

Reported determinism claim. The source manuscript states that decoding uses temperature 0 with *remasking=low_confidence* and that token selection is deterministic conditional on the checkpoint, seed, question assignment, canvas, and NFE schedule. It consequently describes $m(r)$, $d(r)$, $\text{sep}(P)$, and spectral values as deterministic properties of a recorded profile, while bootstrap/permutation summaries carry resampling randomness. The checkpoint fingerprint, exact inputs, implementation, and event schema are absent, so this claim cannot be verified here; cross-seed variation is not claimed.

Reported environment. The source records Python 3.11.13, NumPy 2.4.6, pandas 2.3.1, PyArrow 21.0.0, PyTorch 2.7.1; the model is a locally registered LLaDA-8B-Instruct-scale masked dLLM, blockwise decoded over GSM8K (Cobbe et al. 2021), a fixed 300-question canvas at 256 positions and 48 total NFE per arm. The checkpoint is a private local artifact (registry per the project evaluation ledger) and is *not* byte-identical to the public LLaDA-8B-Instruct release. No checkpoint fingerprint or registry record is available, so release equivalence and the reported scale cannot be checked.

Reported cross-environment replay. The inherited manuscript reports a bit-for-bit replay of the three canvas-256 arm vectors in a second software environment (Python 3.12.3, NumPy 2.5.2, pandas 3.0.5). The raw events, both program versions, outputs, and full digests are absent. This revision did not rerun that analysis and cannot verify the replay; it is not presented as new or independent evidence.

Reproduction boundary. The source reports a fixed seed and a contiguous 300-question development subset, but their searchable internal identifiers are omitted from blind prose and the private crosswalk is unavailable. The current package does not ship event streams, executable programs, registrations, machine-readable results, or the private checkpoint and hence does not support a numeric rebuild. No numeric rebuild was executed for this revision. A reviewer-safe supplement remains an unresolved external requirement.

Execution accounting	Recorded value
Hardware	eight GPUs
Timing	330 running $s = 0.73$ GPU-h; 130 queueing s reported separately and excluded
Shard completion	$L = 16$: 4/4; $L = 64$: 4/4
Event-row check	$L = 16$: 153,600 = 300×512 ; $L = 64$: 499,200 = 300×1664 ; $L = 32$ control: 268,800 = 300×896 rows for the reused contiguous subset of the frozen NFE-48 pool

Table 14: Source-reported execution cost and completion checks (SR/CV) for the two main arms and reused control subset. These values cannot be checked from the current package; the rows describe the inherited report rather than a run made for this revision.

AI-assisted preparation disclosure. The inherited source labels qualitative Figure 1 as image-generated, but no model, prompt, or generation record is present in the accessible package; only its non-evidentiary role can be checked. No image generation was used in this revision. Figure 2 is labeled data-derived by the source, but its data and generator are absent. An AI assistant supported language editing, LaTeX refactoring, and internal review. All empirical values remain explicitly SR/CV rather than being represented as AI-verified or rerun; the authors retain responsibility for scientific content.

C NOTATION AND TERMINOLOGY

Table 15 collects the estimator notation and verdict vocabulary; Table 16 collects the bed names, the treatment variable, and the content-layer terms. Each entry names the section that defines it.

Table 15: Estimator notation and legacy decision-table vocabulary; terms point to their operative definition and do not attribute mechanism.

Term	Meaning (defining location)
$\ell_{i,r}$	Identity-lock time: earliest recorded step after which decode i 's token at relative position r never changes again (Eq. (1)).
$m(r), d(r)$	Per-position median lock time over decodes, and its adjacent difference $m(r+1) - m(r)$ (Section 2).
$\mathcal{B}_P, \mathcal{I}_P$	Candidate- P boundary set $\{r : r \bmod P = P - 1\}$ and its complement, the candidate interior (Section 2).
$\text{sep}(P), P^*$	Min-max period separation $\min_{\mathcal{B}_P} d - \max_{\mathcal{I}_P} d$, and its maximizing candidate (Eq. (2)).
SELECTED	$\text{sep}(P^*) > 0$, bootstrap modal period $= P^*$, mode fraction ≥ 0.95 (registered operational criterion, not a null-calibrated test; Section 2).
REFUTED / NOT-SUPPORTED	Measured $\text{sep}(P) < 0$ for a candidate that the decision table treated as refutable / for the originally registered intrinsic-32 conjunct, whose branch was withdrawn post-data (Table 6).
A-PRIORI layer	What follows from the estimator's algebra alone, before data — e.g. boundary-set nesting (Section 2).
FROM-EVENTS layer	What follows only once measured magnitudes exist — e.g. the sign of $\text{sep}(32)$ (Section 2).
Strictly L -supported	Recorded profile with non-degenerate jumps on all L -block boundaries such that no L -multiple's exclusive-boundary residual outscores L 's boundary minimum (Section 3.1).
$G(P)$, perm. q_{95} , fam. q_{95}	Normalized DFT magnitude of $d(r)$ at frequency $1/P$; its per-candidate permutation 95th percentile; the family-wise version over all candidates (Section A.1).

Term	Meaning (defining location)
L16 arm, L64 arm	The two new matched treatment arms at block length 16 and 64, NFE 48 (Table 1).
L32/NFE48 control	Frozen matched positive control at block length 32, NFE 48: a registered contiguous development subset, $n=300$, with opaque internal identifiers (Sections 2 and 3.1).
L32/NFE32 content bed	Separate, differently budgeted bed used only for the content-association diagnostic (Section A.7).
Partition-plus-required allocation	The composite treatment: changing the block partition necessarily changes the per-block step allocation at fixed total NFE (Table 1).
Measurement-class label	PERIOD-FOLLOWS-BLOCK-LENGTH: a post-data label stating that the selected boundary period follows block length on the tested bed under the active-block recording window; not a mechanism attribution (Sections 1 and 3.1).
BIN8	Restricted null that permutes residualized confidence within 8 equally populated offset-rank bins, preserving offset structure to 4-position granularity (Sections 2 and A.7).
conf0	Position- r initial confidence at block-commit: the model's confidence for r at the first recorded step of its block (Section 2).
Block-unit; resolved position	One decode-block of the content bed; a position whose lock time is defined and that never flips after lock. Terminal-resolved units, which present no within-block pairwise order, are excluded (Table 11).
τ, τ_{res}	Median over units of the per-unit rank association between lock time and confidence; the offset-residualized (content-layer) version (Eq. (3)).

Table 16: Bed, treatment, and content-layer terms; the matched setting and separate content bed remain distinct.